

FORECASTING DAILY SPOT FOREIGN EXCHANGE RATES USING WAVELETS AND NEURAL NETWORKS

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OVERVIEW OF THE TALK

- **Introduction**
 - **What are exchange rates**
 - **Exchange rate dynamics**
 - **Importance of exchange rate forecasting**
- **Conventional methods of exchange rate forecasting**
 - **Economic fundamental models**
 - **Technical models**
- **Wavelets and use of wavelets in the context of the problem**
- **Proposed method and empirical studies**

INTRODUCTION

What are exchange rates?

Nominal exchange rate: Value of domestic currency in terms of a unit of foreign currency

Real exchange rate: Price adjusted nominal exchange rate

Type of nominal exchange rates

Spot rate: price of the currency in the spot market

Forward rate: price of the currency in the forward market

EXCHANGE RATE REGIMES

Floating Exchange Rate

- Nominal exchange is determined by demand and supply
- Free movement of the currency rates

Fixed Exchange Rates

- Nominal exchange rate is set by one country
- Economic fundamentals play a crucial role in fixing exchange rates
- Country's government supports the exchange rate by adjusting foreign exchange reserves-exchange rate intervention policy

EXCHANGE RATE FORECASTING

- **Important and fundamental problem in finance**
- **Challenging area of research to applied statisticians, financial analysts and econometricians**

Importance of exchange rate forecasting

Central Bank- Exchange rate stability

Financial Institutions- trading in currency market

Foreign Institutional Investors, Corporate firms

Why forecasting exchange rate is difficult?

Efficient Market Hypothesis

Louis Bachelier (1900)-*The Theory of Speculation*

Paul Samuelson (1965), Eugene Fama (1970)

- Financial markets are informationally efficient
- Prices on traded financial assets reflect all known information, the collective beliefs of all investors about future prospects
- It is not possible to consistently outperform the market by using any information that the market already knows

Best Forecast: naïve random walk forecasts!

EXCHANGE RATE FORECASTING MODELS

Structural Econometric Models

- Based on analysis of macroeconomic variables that are likely to influence the currency. e.g. relative inflation, interest rates, national income growth, changes in money supply, balance of payments
- Based on theoretical relationships such as *PPP*
- Econometric modeling
- Useful in particular for longer forecasting horizons, fails for short term forecasts

SHORT TERM EXCHANGE RATE MODELS

Technical models

- **Financial Technical Indicator based trigger models**
- **Linear Time Series Models: Auto Regressive (AR), Moving Average (MA), Auto Regressive Integrated Moving Average (ARIMA)**
- **Non-Linear Time Series Models: Bi-Linear, Threshold AR, SETAR, Auto Regressive Conditional Heteroskedastic (ARCH), Generalized ARCH (GARCH)**

For daily spot exchange rate data, most of these models fail to beat the Random Walk!

Alternate Technical Models: Artificial Intelligence Models

Models based on Artificial Neural Networks

Refenes (1996)-hourly spot data

Weigend et. al. (1992)- daily spot data

Hann and Steurer (1996)-weekly data

Lisi and Schiavo (1999)-monthly data

Models based on Genetically Optimized Neural Networks

Nag and Mitra (2002)-daily spot data

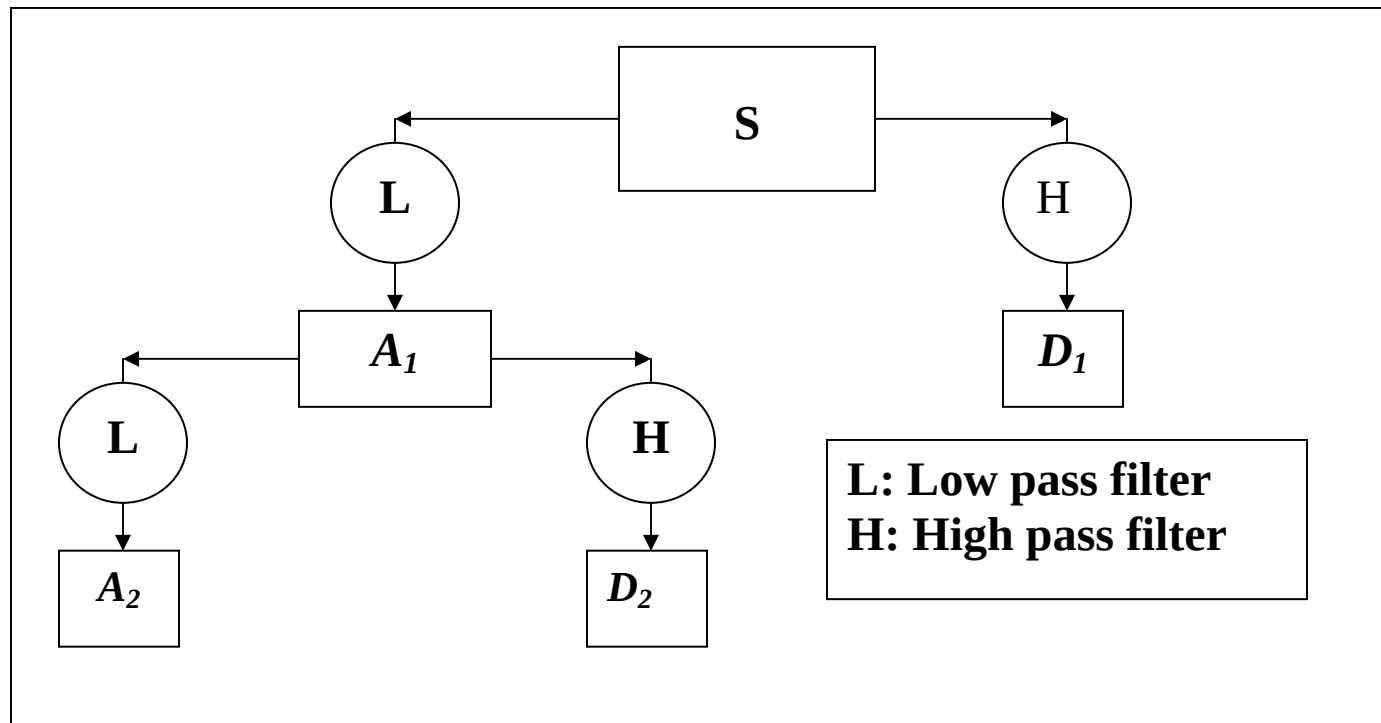
Models outperforms non-linear statistical models and beats RW

WAVELETS IN THE CONTEXT OF THE PROBLEM

Wavelets

- **Wavelet theory has its roots in the classical Fourier analysis**
- **Wavelet analysis is a refinement of the Fourier analysis**
- **Wavelets are defined over a finite domain and unlike the Fourier transform; they are localized both in time and in scale**
- **Ideal for analyzing non-stationary signals and those with transients or singularities**
- **Advantage over traditional Fourier methods in analyzing physical situations where the signals contain discontinuities and sharp spikes**

Schematic Representation of Wavelet Decomposition using Mallat's Pyramid Algorithm



A 2-level wavelet decomposition

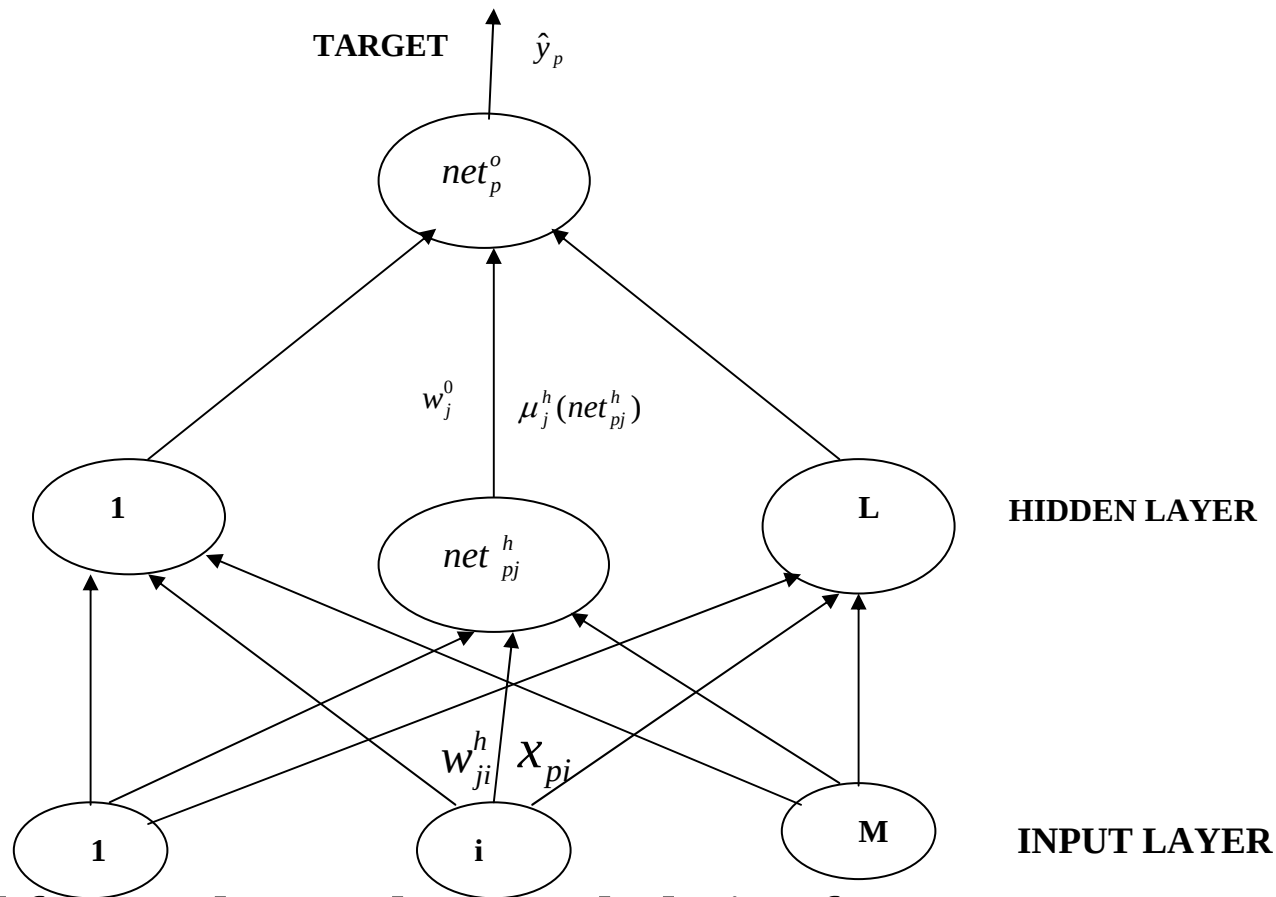
USE OF WAVELETS IN EXCHANGE RATE FORECASTING

- **Observed exchange rate series can be thought of as a mixture of some distinct process components at different scales and volatility levels, which is typical of financial time series**
- **The observed time series is a mixture of such complex processes**
- **Analyst, who is unable to identify the separate scale-related components of the series, is unable to produce models capable of giving accurate forecasts**
- **If we are able to decompose the original time series into scale of resolution related components and model each component separately, we can produce more accurate models**

PROPOSED FORECASTING TECHNIQUE

- **Decompose original exchange rate series using wavelet decomposition and obtain the corresponding approximation and details series at a predetermined level of resolution**
- **Design neural network predictive models for each of the decomposed components of the original series**
- **Input variables of the neural network models for each of the decomposed series, comprise of technical indicators**
- **We further use a generational genetic algorithm with elitism for arriving at the optimum values of the neural network design parameters**

THE ANN STRUCTURE



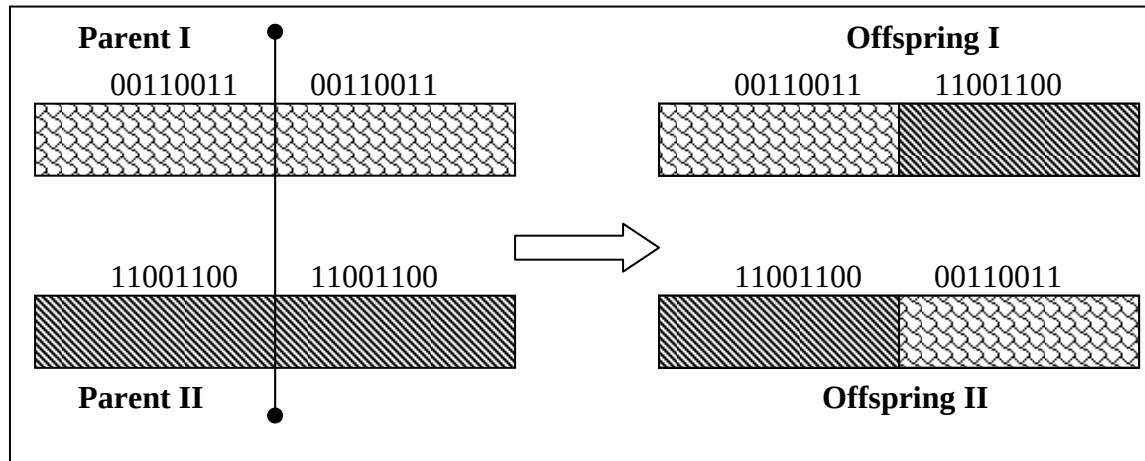
Feed forward neural network design for component models

GENERATIONAL GA WITH ELITISM FOR EACH COMPONENT ANN MODEL

- **Initialization:** Create an initial population from possible inputs variables and network architectures selected at random. Initial population members are transformed to binary coded chromosomes.
- **Fitness Evaluation:** Training and testing these networks using Back Propagation to determine how fit they are for solving the problem. Calculate the fitness of each trained network in the

current population using ranking based approach and preserve the information about the string with highest fitness value.

- **Selection:** Using a stochastic sampling with replacement, populate fit parents pool, size of the pool depending on the generation gap.
- **Crossover:** From the selected parents pool, we select pairs in order and apply a 2-point crossover (with a pre-assigned crossover probability), exchanging genetic material of parents to obtain offspring chromosomes.



A one-point crossover

- **Mutation:** Apply mutation on the new chromosome strings with small pre-assigned mutation probability.
- **Elitism:** Use elitist strategy to fill the generation gap.
- Repeat the steps 2 to 6 till the convergence criterion is reached.

EMPIRICAL STUDIES

Modeling of daily spot exchange rate data

Australian Dollar/US Dollar, Canadian Dollar/US Dollar, Japanese Yen/US Dollar, US Dollar/Pound Sterling, French Franc/US Dollar, Swiss Franc/US Dollar, Dutch Guilders/US Dollar and German Deutsch Mark/US Dollar

Data Source: Reuters

One-step ahead forecasting models: Target variable is the closing exchange rate one day ahead

Multi-step ahead models: Target variable is the closing spot rate at the chosen lead period

Sample size

1000 data points, stretched over a period of three and half years.

Data Period: 2004-2007

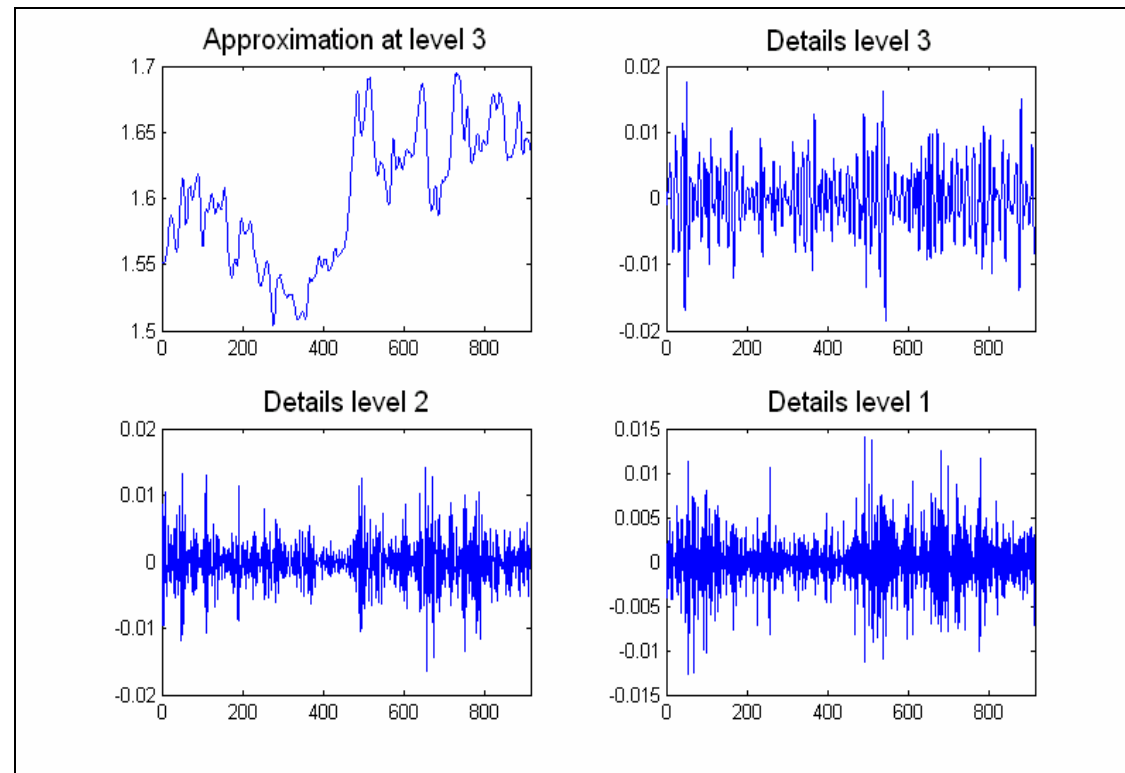
Data Splitting

Last 10% data points (test set data) are reserved for evaluation of the out of sample performance and are not used during the model building stage with training set (first 90%).

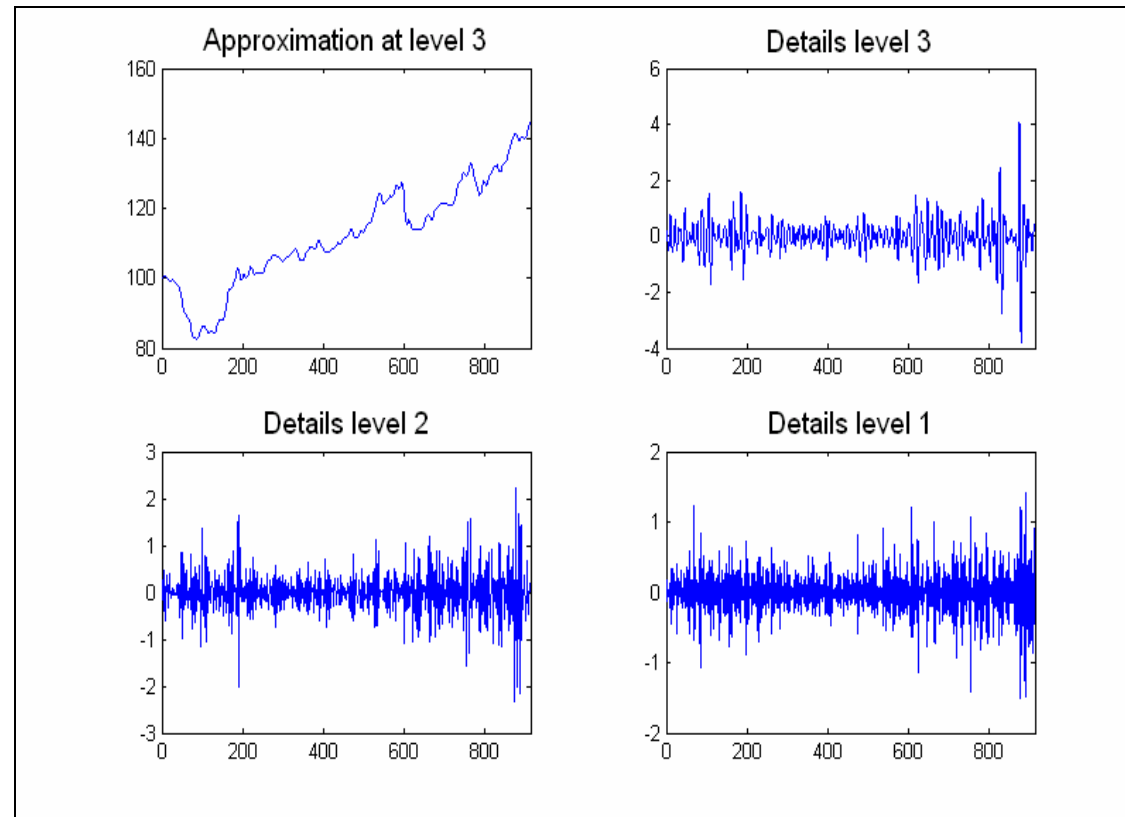
Wavelet Decomposition

Using a Daubechies-5 wavelets, we obtain a wavelet decomposition of the ‘training set’ data

Daubechis-5 3-level decomposition of US Dollar/Pound Sterling rate



Daubechis-5 3-level decomposition of Japanese Yen /US Dollar rate



Component Modeling

Genetically optimized neural networks for modeling each component

Exchange rate Forecasts

The component-wise forecasts are combined to get the forecasts of the original series

Out-of-Sample Testing

After the model building with the ‘training set’ data using the proposed methodology is done, we use the respective component models to generate forecasts for the ‘test data’ set

Performance Measures

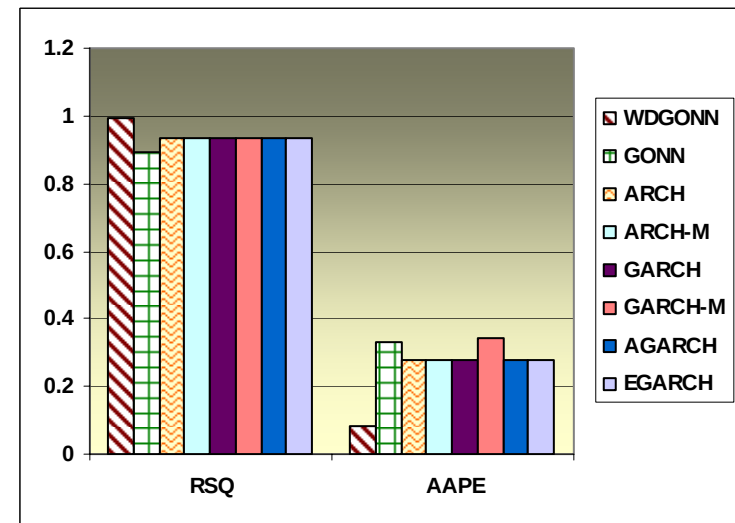
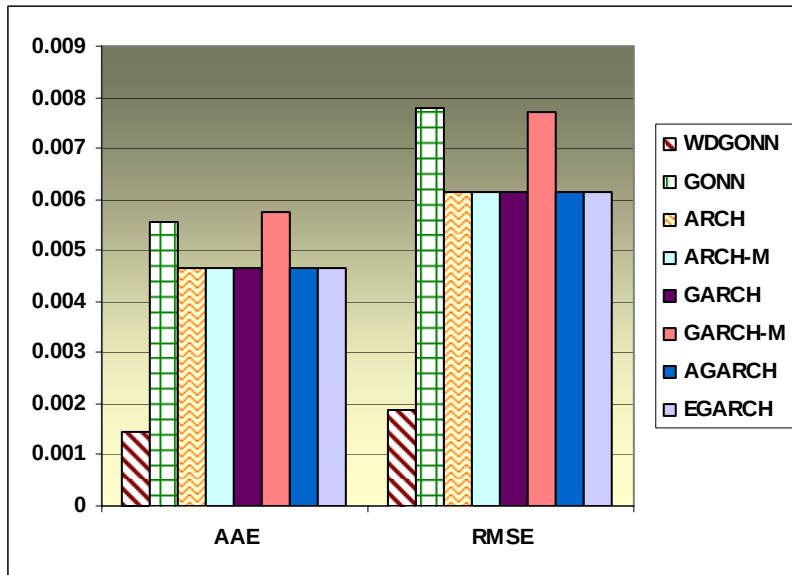
- **Average Absolute Error (AAE)**
- **Average Absolute Percentage Error (AAPE)**
- **Root Mean Square Error (RMSE)**
- **Percentage of Correct Movements (PCM)**
- **R-Square (RSQ)**

Models Compared

- **Proposed Model (WDGONN)**
- **ARCH, GARCH, AGARCH, EGARCH Models**
- **Ordinary Genetically optimized ANN models (GONN)**

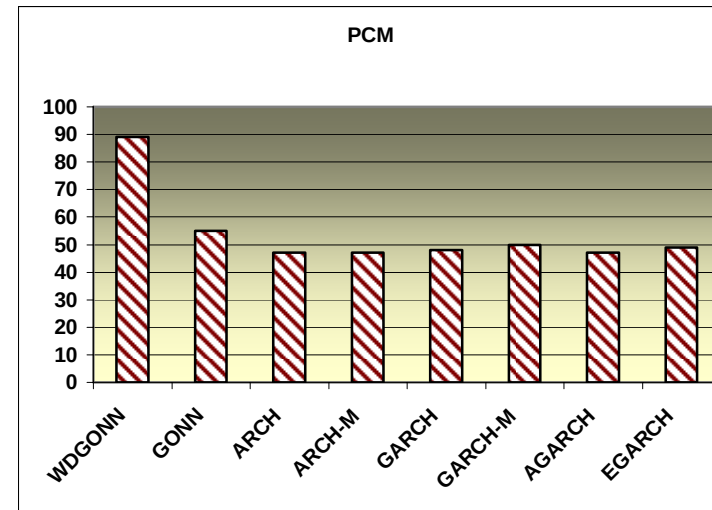
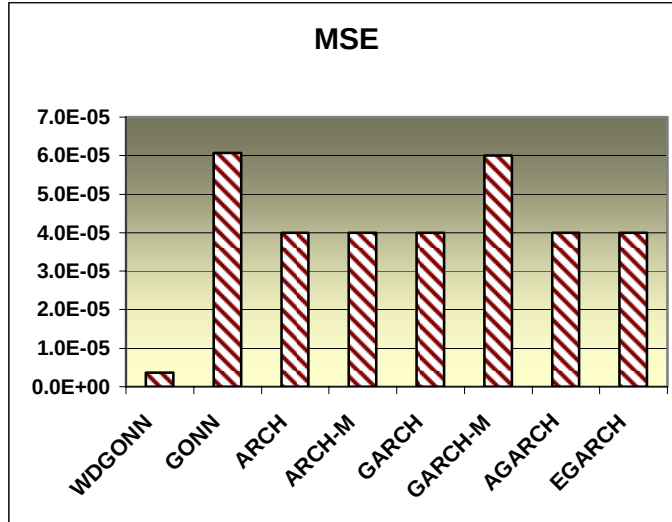
OUT-OF-SAMPLE PERFORMANCE RESULTS

One-step ahead prediction: US Dollar/Pound Sterling



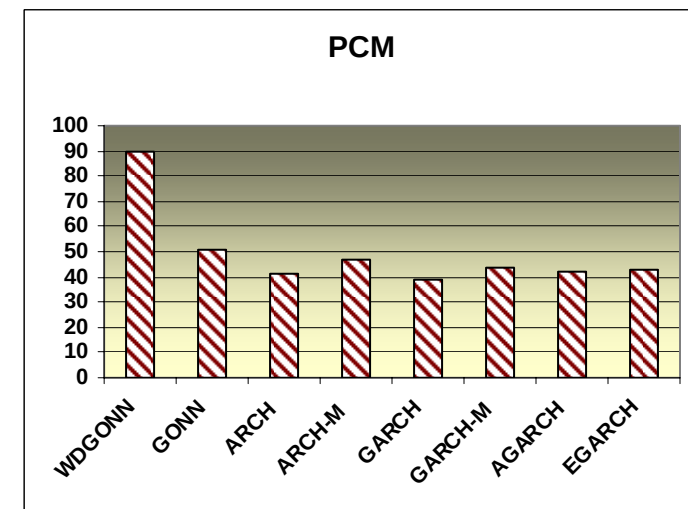
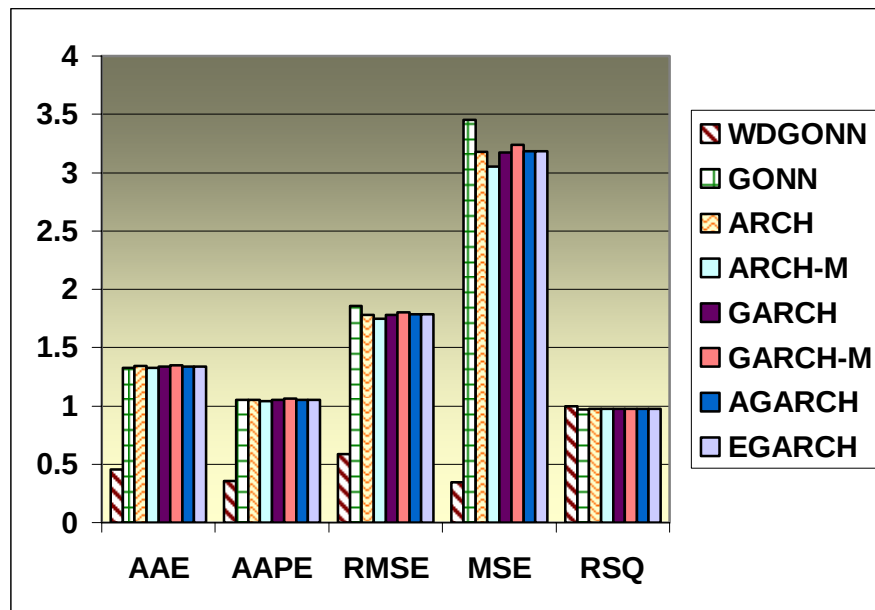
OUT-OF-SAMPLE PERFORMANCE RESULTS

One-step ahead prediction: US Dollar/Pound Sterling



OUT-OF-SAMPLE PERFORMANCE RESULTS

One-step ahead prediction: Japanese Yen /US Dollar



Out-of-sample performance at different levels

Results For Japanese Yen/US Dollar

Component	AAE	MSE	RSQ
Level 3 Approximation	0.37441	0.28402	0.99744
Level 3 Detail	0.11790	0.02451	0.98223
Level 2 Detail	0.12016	0.02692	0.95349
Level 1 Detail	0.15881	0.04953	0.88863

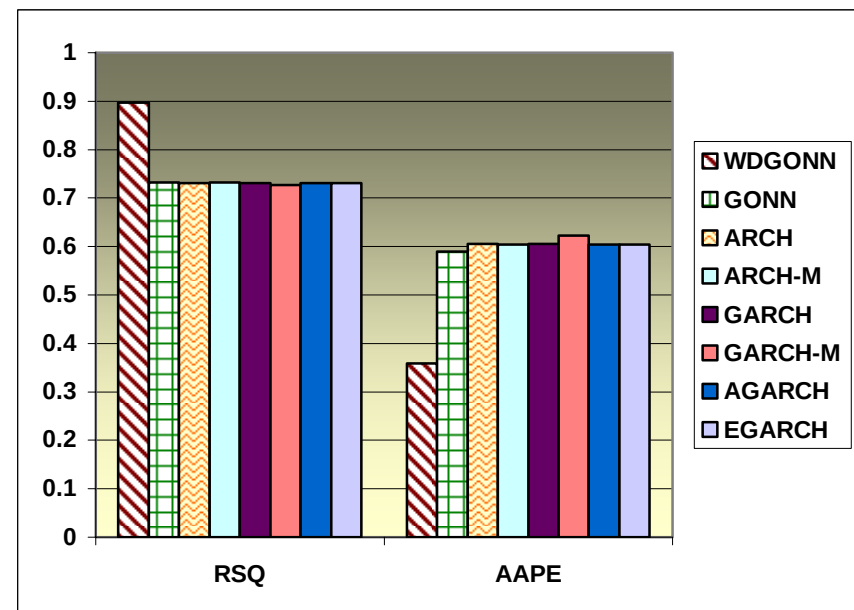
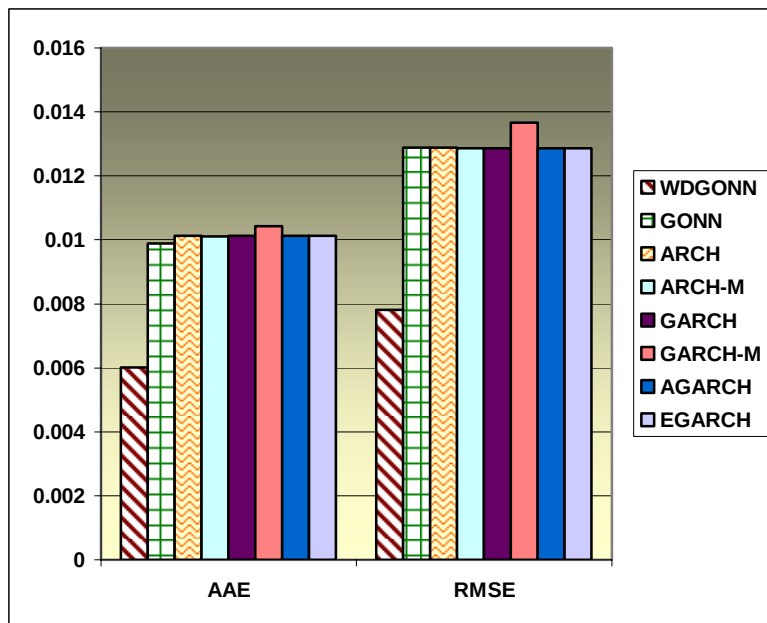
Observation

Among the detail series, the coarsest series ‘level 3 detail’, which is the least volatile component, is easiest to forecast.

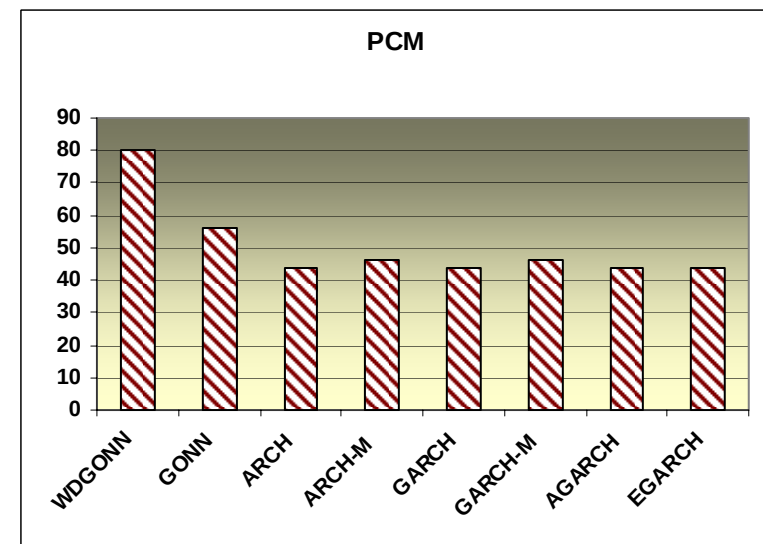
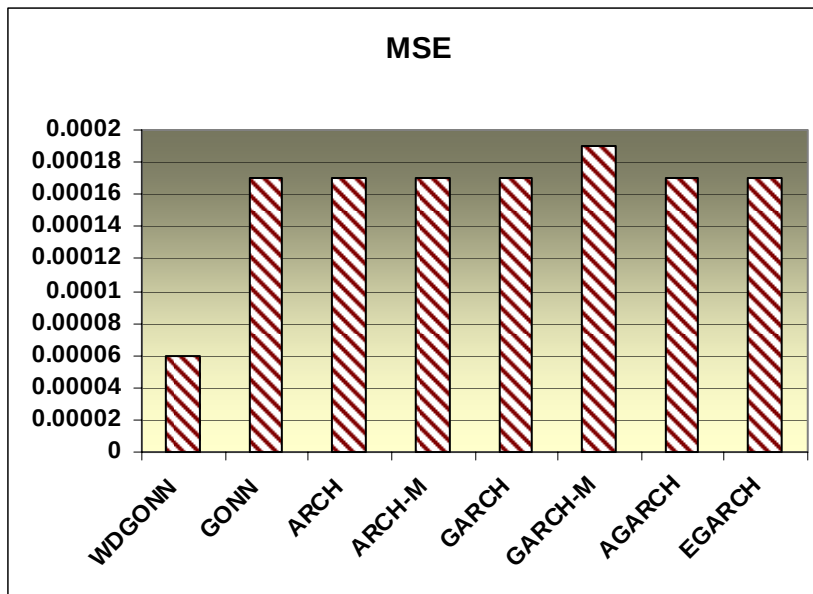
Multi-step ahead prediction

- Performance deteriorate as lead period increases

4-step ahead US Dollar/Pound Sterling



4-step ahead US Dollar/Pound Sterling



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